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BOARD DIPLOMA EXAMINATION, (C-20)
MARCH/APRIL—2025
DCE – FIFTH SEMESTER EXAMINATION
ENVIRONMENTAL ENGINEERING

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. State the following terms :
 - (a) Acid rain
 - (b) Global warming
2. State any three objectives of treatment of water.
3. State any six water borne diseases.
4. Discuss the terms (a) water main and (b) gooseneck.
5. State the different shapes of surface drains.
6. List any six sewer appurtenances.
7. List any three functions of grit chambers.
8. State any three methods of solid waste disposal.
9. State the objectives of disinfection of wells.
10. State the effects of air pollution.

PART—B

8×5=40

- Instructions :** (1) Answer either (a) **or** (b) from each question.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** (a) Predict the population of a town for the years 2011, 2021 from the following census data by Arithmetic increase method, Geometric increase method and Incremental Increase method : 8

Year	1961	1971	1981	1991	2001
Population	39100	46000	58500	73000	90600

(OR)

- (b) List merits and demerits of Stoneware, Cast iron and Cement concrete pipes used for conveyance of water. 8

- 12.** (a) Explain the advantages and disadvantages of Intermittent and Continuous systems of water supply. 8

(OR)

- (b) Explain briefly the accessories and functions of an Overhead tank. 8

- 13.** (a) Explain briefly the different methods of Sewage collection and disposal. 8

(OR)

- (b) (i) State any five factors affecting stormwater flow. 5
(ii) State Chezy's and Manning's formulae. 3

- 14.** (a) With the help of a neat sketch explain briefly the construction and working of trickling filter. 8

(OR)

- (b) Design a Septic tank for a small colony of 200 persons with an average water supply of 150 lpcd. 8

15. (a) Briefly explain any three different types of solid waste disposal. 8

(OR)

(b) Draw a neat sketch showing drainage arrangement for a building. 8

PART—C

10×1=10

Instructions : (1) Answer the following question.
(2) The question carries **ten** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

16. (a) Briefly explain the methods of control of Air pollution. 6

(b) State the objectives of chlorination of sewage effluents. 4

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